

1. Administrative & Study Identification

Full Study Title: A Modular Phase I/II Open-label, Multicenter Study to Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, Immunogenicity, and Efficacy of AZD4512 Monotherapy or in Combination With Other Anticancer Agent(s), in Participants With Relapsed/Refractory B-cell Non-Hodgkin Lymphoma (B-NHL) **Short Title/Acronym:** Lumi-NHL **Protocol Number:** D9890C00001 **NCT Number:** NCT07123454 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD4512 (CD22-targeted antibody-drug conjugate) **Phase of Development:** Phase I/II **Medical Monitor:** [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety, determine the Maximum Tolerated Dose (MTD), and/or recommended dose for continued evaluation of AZD4512 monotherapy or in combination with other anticancer agents. **Secondary Objectives:** To evaluate the objective response rate (ORR), complete response (CR), duration of response (DoR), progression-free survival (PFS), overall survival (OS), pharmacokinetics, and immunogenicity.

2.2 Background & Rationale To address the unmet medical need in patients with relapsed or refractory B-cell Non-Hodgkin Lymphoma (R/R B-NHL) who have exhausted at least two prior lines of therapy and lack additional standard options with a survival benefit. AZD4512 provides a novel CD22-targeted antibody-drug conjugate approach to overcome treatment resistance in these patient populations.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Modular dose escalation and expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 91 participants **Number of Sites:** Global and Regional breakdown (Multicenter) **Estimated Study Start:** 24-Sep-2025 **Estimated Primary Completion:** 01-Mar-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years with documented diagnosis of B-cell non-Hodgkin lymphoma (B-NHL) as per World Health Organization (WHO) 2021 classification. 2. Relapsed or refractory disease after at least two prior lines of therapy: Large B-cell Lymphoma (LBCL) patients must have progressed after anti-CD20 and ≥ 1 systemic chemotherapy regimen, and be ineligible/considered for CAR-T; Mantle cell lymphoma (MCL) patients must have received an anti-CD20 and a BTK inhibitor; Follicular lymphoma (FL) patients must have failed an anti-CD20 therapy. 3. Measurable disease by Lugano 2014 criteria, Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 , and adequate organ and bone marrow function.

4.2 Exclusion Criteria 1. Diagnosis of post-transplant lymphoproliferative disease, Richter's transformation, Burkitt's lymphoma, chronic lymphocytic leukemia (CLL) / small lymphocytic lymphoma (SLL), or active Central nervous system (CNS) involvement from B-NHL. 2. Prior Chimeric antigen receptor-T (CAR-T) or T cell engager therapies within 90 days prior to Cycle 1 Day 1 (C1D1), or other investigational/systemic anticancer therapies within 21 days or 5 half-lives. 3. Allogeneic Hematopoietic stem cell transplantation (HSCT) within 180 days or autologous HSCT within 90 days prior to study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Modular dose escalation and expansion of AZD4512. Dosing specific to protocol cohorts. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Treatment will continue until disease progression, unacceptable toxicity, or withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Localized palliative radiotherapy, low-dose corticosteroids. **Prohibited:** Strong CYP3A4 inhibitors, corrected QT interval-prolonging agents at Cycle 1, and active curative radiation within 14 days of C1D1.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, Baseline Scans, Tumor biopsies, Baseline Labs
Baseline	Day 1	First Dose of AZD4512, Vital Signs, Safety Monitoring
Interim	Regular Cycles	Safety Labs, Efficacy Assessment (Tumor Scans), PK/PD monitoring
End of Study	Up to 30 Days Post-Dose	Final Vital Signs, Exit Interview, IP Accountability, Adverse Event follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability measures including the frequency, duration, and severity of treatment-emergent adverse events (TEAEs), treatment-related adverse events (TRAEs), serious adverse events (SAEs), and dose-limiting toxicities (DLTs) to determine MTD. **Measurement Method:** Clinical observations, physical examinations, and laboratory analysis evaluated up to 30 (+7) days after the last dose.

7.2 Secondary & Exploratory Endpoints * **Endpoint 1:** Objective Response Rate (ORR), Complete Response (CR), and Duration of Response (DoR). * **Endpoint 2:** Progression-Free Survival (PFS) and Overall Survival (OS). * **Endpoint 3:** Pharmacokinetics (PK) and Pharmacodynamics (PD). * **Endpoint 4:** Immunogenicity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via ongoing clinical site visits and laboratory monitoring. **Serious Adverse Events (SAEs):** Must be reported within typically 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External Safety Review Committee (SRC) to review dose escalation and boundary clearing parameters.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy, Safety Set for adverse events. **Sample Size Justification:** N=91\$ participants. Appropriately scaled for a Phase I/II dose escalation and expansion study to determine MTD boundaries and capture early efficacy signals. **Handling of Missing Data:** Standard imputation rules applicable for missing continuous laboratory and survival endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard periodic onsite and remote monitoring visits to ensure protocol adherence. **Audit Trail:** eCRF locking, source data verification, and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: D9890C00001 **Registry Number:** NCT07123454 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Lumi-NHL **Official**

Regulatory Title: A Modular Phase I/II Open-label, Multicenter Study to Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, Immunogenicity, and Efficacy of AZD4512 Monotherapy or in Combination With Other Anticancer Agent(s), in Participants With Relapsed/Refractory B-cell Non-Hodgkin Lymphoma (B-NHL) (Lumi-NHL)

12. Clinical Framework (Study Specific)

Primary Condition: Relapsed/Refractory B-cell Non-Hodgkin Lymphoma (B-NHL) **Diagnosis Threshold:** Histological confirmation of disease progression following ≥ 2 prior lines of systemic therapy **Medical Methodology:** Targeted CD22 antibody-drug conjugate (ADC) administration **Optimization Target:** Maximum Tolerated Dose (MTD) definition and anti-tumor Objective Response

13. Study Procedures & Methodology

Management Pathway: Sequential modular dose escalation followed by dose expansion **Education Protocol (HCP):** Investigator Meeting and Oncology Protocol Site Initiation Visits **Education Protocol (Patient):** Informed consent overview and treatment timeline counseling **Quality Audit Mechanism:** Site monitoring, safety review committee checks, and central data review

14. Observation & Follow-Up Schedule

Total Duration: Expected up to March 2028 (or until disease progression/unacceptable toxicity) **Onsite Visit Cadence:** Per standardized oncology treatment cycles **Remote Monitoring Frequency:** Ongoing centralized data review **Checklist Review Interval:** Safety tracking at the end of each dose escalation cohort

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				